

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is a DoS attack.

The logs show that: an unfamiliar volume of incoming traffic, which causes the web server not to respond and gives a timeout error message

This event could be a type of DoS attack called SYN flooding.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. The [SYN] packet is the initial request from an employee visitor trying to connect to a web page hosted on the web server. SYN stands for "synchronise."
2. The [SYN, ACK] packet is the web server's response to the visitor's request, agreeing to the connection. The server will reserve system resources for the final step of the handshake. SYN, ACK stands for "synchronise acknowledge."
3. The [ACK] packet is the visitor's machine acknowledging permission to connect. This is the final step required to make a successful TCP connection. ACK stands for "acknowledge."

Explain what happens when a malicious actor sends a large number of SYN packets all at once: When a malicious actor sends a large number of SYN packets that are bigger than the maximum size of the ICMP, it overwhelms the website server and causes it to slow down and eventually shut down

Explain what the logs indicate and how that affects the server: The logs indicated that the web server stopped responding to legitimate employee visitor traffic. The server stopped responding and was down